

The empirical recurrence for OEIS A297611: recovering a conjecture that is not written down

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2 September 2026

Abstract

OEIS A297611 records “Empirical recurrence of order 96”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 178$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 96, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 96 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

2020 Mathematics Subject Classification. 05A15, 11B37, 15A18.

1 A conjecture one cannot read

OEIS A297611 is “Number of $5 \times n$ 0..1 arrays with every 1 horizontally, diagonally or antidiagonally adjacent to 0, 1 or 3 neighboring 1s.”. It has offset 1 and begins

32, 327, 4997, 52512, 654033, 7847849, 95770934, 1148443249, ...

The entry states:

Empirical recurrence of order 96 (see link above)

As of the “Last modified” line on the live entry (Jan 01 2018 14:57:09, revision 5) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 178$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 178$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S' = 356$ exact terms of the model returns order 96 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{96} c_i a(n-i),$$

c_1	4	c_{25}	3439586796068	c_{49}	−371424893043251059	c_{73}	−17420911257497250
c_2	58	c_{26}	−17201915371515	c_{50}	15707786907511148	c_{74}	2592259625802600
c_3	249	c_{27}	52555252186229	c_{51}	56827265625617026	c_{75}	−3276990162421202
c_4	2978	c_{28}	−180790783626878	c_{52}	13849505718518046	c_{76}	3425918885543202
c_5	1984	c_{29}	−91089497482272	c_{53}	596172178726224928	c_{77}	7534728774825302
c_6	−43940	c_{30}	422634565757370	c_{54}	−104303948282844401	c_{78}	−2273325712486858
c_7	−84123	c_{31}	−311556116336634	c_{55}	−118369637239086193	c_{79}	36292103726468
c_8	−1058906	c_{32}	1025148212407216	c_{56}	−89922158523897408	c_{80}	10151502685860
c_9	−698120	c_{33}	513682060638487	c_{57}	−570003773053835610	c_{81}	−975222229599236
c_{10}	17951127	c_{34}	−3398366308359600	c_{58}	205539310939881388	c_{82}	320696576730292
c_{11}	10201269	c_{35}	1060605321374379	c_{59}	124407772384884541	c_{83}	−401083690193972
c_{12}	68374001	c_{36}	−3856724842421664	c_{60}	3771812295666376	c_{84}	147409178271720
c_{13}	142323241	c_{37}	1349833356662928	c_{61}	368151605255210599	c_{85}	66174277818552
c_{14}	−2411547952	c_{38}	13840369036899333	c_{62}	−165740086561984029	c_{86}	−31948299604416
c_{15}	−1141931466	c_{39}	−3676538479913306	c_{63}	−60130697482285089	c_{87}	35898128995680
c_{16}	6387182724	c_{40}	14148160327241220	c_{64}	71511214092769579	c_{88}	−18727915171712
c_{17}	−13176851729	c_{41}	−28093566692464324	c_{65}	−234663394139052920	c_{89}	4604522669504
c_{18}	104730242225	c_{42}	−30595954827375440	c_{66}	55122643153168180	c_{90}	−1335524299008
c_{19}	115083731050	c_{43}	12174174370317699	c_{67}	49104712064272792	c_{91}	273858822144
c_{20}	−614161929665	c_{44}	−38938480562793881	c_{68}	−29750224767291164	c_{92}	−45492656640
c_{21}	270331405116	c_{45}	139170114169321335	c_{69}	91364920060213842	c_{93}	−1612708864
c_{22}	−1194360763753	c_{46}	29748002079710414	c_{70}	−28780232040998952	c_{94}	1065347072
c_{23}	−4166279251111	c_{47}	−25724585745491760	c_{71}	−16281488987210408	c_{95}	−80728064
c_{24}	16535398816123	c_{48}	48937188730674281	c_{72}	13923418570942242	c_{96}	−2211840

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 96, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $96 + 4$ terms removed returns order 96 as well, so $\deg q_{\min} = 96 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 16 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $96 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 96 that this sequence satisfies — and that family turns out to have exactly one member.

References

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